Senior Data Platform Engineer | Shanghai, China | Open to Global Remote | jromeguan@gmail.com | https://github.com/jlty258

Guan Chen

Summary

Platform engineer with 13+ years building distributed data infrastructure. Designed PB-scale analytical platforms, streaming systems processing 10B+ events/day across 1M+ connected vehicles, and privacy computing execution pipelines with query planning and storage abstraction.

Experience

Senior Data Platform Engineer

Shanghai Puxin Future Internet Research Institute | Dec 2024 – Present

- Designed PQL execution planning pipeline converting declarative queries into optimized distributed DAGs across heterogeneous compute backends
- Architected query planner with operator rewriting and cost-aware execution graph construction, separating query semantics from execution strategy
- Built reusable Data Service layer with storage abstraction interfaces, decoupling execution engines from physical storage implementations
- Established platform interfaces and engineering standards across planner, scheduler, execution engine, and data service components
- Built benchmark platform providing measurable evidence for execution planning and runtime optimization decisions

Senior Data Platform Engineer / Technical Architect

Shanghai Jiayu Intelligent Robotics | Jun 2020 – Sep 2024

- Architected streaming platform ingesting 10B+ telemetry events/day from 1M+ connected vehicles on a 1000+ vCore Flink cluster
- Designed Kafka ingestion layer decoupling vehicle T-Box producers from downstream stream processing and storage consumers
- Built Flink processing pipelines for real-time aggregation and enrichment, with Apache Doris serving as the real-time warehouse layer
- Defined metadata models and data governance standards across streaming datasets, enabling schema discovery and data quality enforcement
- Integrated Dinky platform for Flink job management and operational workflows across the streaming cluster

Senior Data Platform Engineer

Ping An Securities | May 2017 – Apr 2020

- Architected enterprise metric platform unifying PB-scale Hive analytics with Elasticsearch serving 100000+ searchable metric fields

- Designed dynamic schema expansion mapping 2000+ Hive physical columns to Elasticsearch indices through a Map-based translation layer
- Optimized Hive and Hadoop workloads on 1000+ vCore cluster, improving query latency and resource utilization across PB-scale storage
- Decoupled analytical computation from interactive search, enabling business applications to query metrics without direct Hive access

Data Platform Engineer

Shanghai Jiayin Data Technology

- Built data processing pipelines for restaurant intelligence and commercial Wi-Fi analytics using Storm, Hive, and Hadoop

Research Engineer

Shanghai Ship and Shipping Research Institute

- Applied Hadoop and Hive to hydrodynamic simulation workloads in scientific computing domain

Skills

Java, Python, C | Kafka, Flink, Spark, Hive, Hadoop | HDFS, Doris, Hudi, Elasticsearch | Linux, Docker, Kubernetes | Streaming, Analytical Computing, Privacy Computing, Metadata, Data Governance, Execution Planning

Education

Master, Naval Architecture and Ocean Engineering — Shanghai Ship and Shipping Research Institute

Bachelor, Hydraulic Engineering — Tianjin University